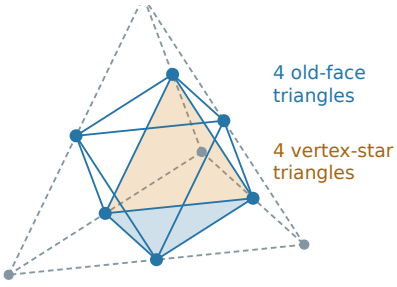
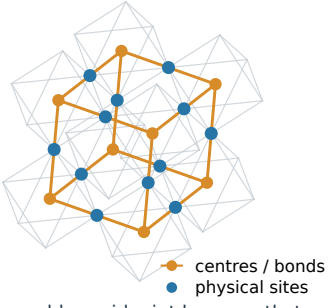
Tetrahedron → inscribed octahedron



Six edge midpoints become the six covering sites.

Corner-sharing octahedra; cubic centres



Every blue midpoint has exactly two owners.